

Skill Arena CTO & Codex Development Charter - Enterprise Edition

Version 1 Engineering Governance Manual

Architecture Principles

All critical business logic executes server-side. Services must be modular, scalable and independently deployable. All critical business logic executes server-side. Services must be modular, scalable and independently deployable. All critical business logic executes server-side. Services must be modular, scalable and independently deployable.

Zero Trust Client

Clients never calculate balances, rewards, rankings, treasury values, trust scores or challenge outcomes. Clients never calculate balances, rewards, rankings, treasury values, trust scores or challenge outcomes. Clients never calculate balances, rewards, rankings, treasury values, trust scores or challenge outcomes.

Backend Standards

Primary backend services implemented in Golang. AI and analytics services implemented in Python. Primary backend services implemented in Golang. AI and analytics services implemented in Python. Primary backend services implemented in Golang. AI and analytics services implemented in Python.

Database Standards

Use PostgreSQL as the system of record. Apply migrations, backups, auditing and transactional integrity controls. Use PostgreSQL as the system of record. Apply migrations, backups, auditing and transactional integrity controls. Use PostgreSQL as the system of record. Apply migrations, backups, auditing and transactional integrity controls.

API Standards

Versioned APIs, authentication, authorization, rate limiting, monitoring and documentation are mandatory. Versioned APIs, authentication, authorization, rate limiting, monitoring and documentation are mandatory. Versioned APIs, authentication, authorization, rate limiting, monitoring and documentation are mandatory.

Security Standards

JWT, refresh tokens, MFA, RBAC, encryption, secrets management and audit logging required. JWT, refresh tokens, MFA, RBAC, encryption, secrets management and audit logging required. JWT, refresh tokens, MFA, RBAC, encryption, secrets management and audit logging required.

Treasury Standards

Every financial event must be ledger-backed, auditable and reversible through controlled workflows. Every financial event must be ledger-backed, auditable and reversible through controlled workflows. Every financial event must be ledger-backed, auditable and reversible through controlled workflows.

Wallet Standards

Wallet balances derive from immutable ledger events rather than mutable counters. Wallet balances derive from immutable ledger events rather than mutable counters. Wallet balances derive from immutable ledger events rather than mutable counters.

Challenge Engine Standards

Maze generation, validation, difficulty calculations and outcome verification remain server-authoritative. Maze generation, validation, difficulty calculations and outcome verification remain server-authoritative. Maze generation, validation, difficulty calculations and outcome verification remain server-authoritative.

Replay Standards

Every competitive event must generate replay records suitable for validation and dispute resolution. Every competitive event must generate replay records suitable for validation and dispute resolution. Every competitive event must generate replay records suitable for validation and dispute resolution.

Anti-Cheat Standards

Behavioral analysis, trust scoring, replay intelligence and verification workflows are mandatory. Behavioral analysis, trust scoring, replay intelligence and verification workflows are mandatory. Behavioral analysis, trust scoring, replay intelligence and verification workflows are mandatory.

DevOps Standards

CI/CD pipelines, staging environments, rollback procedures and infrastructure monitoring are required. CI/CD pipelines, staging environments, rollback procedures and infrastructure monitoring are required. CI/CD pipelines, staging environments, rollback procedures and infrastructure monitoring are required.

Testing Standards

Unit testing, integration testing, load testing, security testing and regression testing required. Unit testing, integration testing, load testing, security testing and regression testing required. Unit testing, integration testing, load testing, security testing and regression testing required.

Monitoring Standards

Track infrastructure health, application health, treasury health and security indicators. Track infrastructure health, application health, treasury health and security indicators. Track infrastructure health, application health, treasury health and security indicators.

Incident Management

Follow detection, containment, investigation, remediation and post-mortem procedures. Follow detection, containment, investigation, remediation and post-mortem procedures. Follow detection, containment, investigation, remediation and post-mortem procedures.

Release Management

Deploy through staged environments with approval gates and rollback plans. Deploy through staged environments with approval gates and rollback plans. Deploy through staged environments with approval gates and rollback plans.

Data Governance

Data ownership, retention policies, auditability and privacy requirements must be enforced. Data ownership, retention policies, auditability and privacy requirements must be enforced. Data ownership, retention policies, auditability and privacy requirements must be enforced.

Scalability Standards

Systems must support future multi-game expansion and horizontal scaling. Systems must support future multi-game expansion and horizontal scaling. Systems must support future multi-game expansion and horizontal scaling.

Documentation Standards

All services, APIs, schemas and operational procedures must be documented. All services, APIs, schemas and operational procedures must be documented. All services, APIs, schemas and operational procedures must be documented.

Success Criteria

Platform remains secure, scalable, maintainable, auditable and fair. Platform remains secure, scalable, maintainable, auditable and fair. Platform remains secure, scalable, maintainable, auditable and fair.